

Chapter 6: Exponentials and Logarithms

A Level Pure Maths Notes · Rexponential

Why this chapter matters

The laws of logarithms, and probably base-10 and base-2 logs specifically, should already be familiar. A level introduces (or leans much more heavily on) the **natural logarithm** $\ln x = \log_e x$ and Euler's number $e \approx 2.71828\dots$, because e has the special property that $d/dx(e^x) = e^x$ — it's the base that makes calculus with exponentials clean, which is exactly why it dominates from here through Chapters 7 and 8. This chapter also covers using logs and exponentials to model real growth and decay processes, and a technique — reducing to linear form — for fitting a model to experimental data using logs.

A reminder before we start: log and exponential functions are **inverses** of each other (a^x and $\log_a x$ undo one another), so almost everything in this chapter can be understood as “what does taking a log, or exponentiating, do to an equation?”

6.1 Laws of logarithms

Concept

A logarithm answers the question “what power do I need?” — $\log_a x = y$ means $a^y = x$. The laws of logarithms are really just the laws of indices translated into log language: since multiplying powers of the same base adds the exponents, multiplying two numbers corresponds to *adding* their logs, and so on.

For any valid base a ($a > 0$, $a \neq 1$) and $x, y > 0$:

LAWS OF LOGARITHMS

- $\log_a(xy) = \log_a x + \log_a y$ (product law)
- $\log_a(x/y) = \log_a x - \log_a y$ (quotient law)
- $\log_a(x^n) = n \log_a x$ (power law)
- $\log_a a = 1$, and $\log_a 1 = 0$, for any valid base a
- $\log_a x = \ln x / \ln a$ (change of base — useful when your calculator only has $\ln$ and $\log_{10}$ buttons)

These laws let you combine several log terms into one, or split one log term into several — which direction is useful depends on the problem: combining is usually the move when *solving* equations (so you can drop the logs), splitting is usually the move when *simplifying* an expression.

Worked example

EXAMPLE 6.1

Solve $\log_2(x) + \log_2(x - 2) = 3$.

Step 1 Combine the left side using the product law: $\log_2(x) + \log_2(x-2) = \log_2[x(x-2)]$ (M1). So $\log_2[x(x-2)] = 3 \Rightarrow x(x-2) = 2^3 = 8$ (M1, converting from log form to index form).

Step 2 Expand: $x^2 - 2x - 8 = 0 \Rightarrow (x-4)(x+2) = 0 \Rightarrow x = 4$ or $x = -2$ (A1).

Step 3 Check domain: the original equation needs $x > 0$ AND $x - 2 > 0$, i.e. $x > 2$. $x = -2$ fails this, so it's rejected.

Answer Only $x = 4$ is a valid solution (A1, with the rejection justified).

EXAM TIP

Solving the resulting equation correctly but forgetting to check both solutions against the domain of the *original* logarithmic expressions — logs of zero or negative numbers are undefined, so any solution that would make an argument ≤ 0 must be discarded, and examiners specifically award a mark for showing you've checked this.

6.2 Solving exponential and logarithmic equations

Concept

Exponential equations (the unknown is in the exponent, like $5^{2x-1} = 30$) are solved by taking logs of both sides, which “brings the exponent down” using the power law. Logarithmic equations (the unknown is inside a log, like $\log_3(2x+1) = 2$) are solved by converting to index form. These are genuinely inverse procedures — recognising which type you're facing tells you immediately which direction to work in.

Procedure

- Exponential equation** $a^{f(x)} = b$: take $\ln$ (or $\log$) of both sides $\Rightarrow f(x) \cdot \ln a = \ln b \Rightarrow f(x) = \ln b / \ln a$, then solve for x .
- Logarithmic equation** $\log_a(f(x)) = b$: rewrite in index form $\Rightarrow f(x) = a^b$, then solve for x , and **always check the solution against the domain** of the original log expression(s).

Worked example

EXAMPLE 6.2A

Solve $5^{2x-1} = 30$, giving your answer to 3 significant figures.

Step 1 Take $\ln$ of both sides: $\ln(5^{2x-1}) = \ln 30 \Rightarrow (2x-1) \ln 5 = \ln 30$ (M1, using the power law to bring the exponent down).

Step 2 $2x - 1 = \ln 30 / \ln 5 = 2.113... \Rightarrow 2x = 3.113... \Rightarrow x = 1.5565...$ (M1).

Answer $x = 1.56$ (3 s.f.) (A1).

EXAMPLE 6.2B

Solve $\log_3(2x+1) = 2$.

Step 1 Convert to index form: $2x + 1 = 3^2 = 9$ (M1).

Step 2 $2x = 8$ (M1).

Answer $x = 4$ (A1). (Quick domain check: $2(4)+1 = 9 > 0$, so this is valid.)

EXAM TIP

Writing $\ln(5^{2x-1})$ as $(2x-1)\ln 5$ correctly, but then dividing incorrectly — e.g. dividing $\ln 30$ by 5 instead of by $\ln 5$. Keep $\ln 5$ and $\ln 30$ as whole quantities throughout the algebra, and only evaluate them numerically at the very last step to minimise rounding errors.

6.3 Graphs of a^x , e^x , and $\ln x$

Concept

The graphs of $y = a^x$ (for $a > 1$) and $y = e^x$ are increasing, always positive, pass through $(0, 1)$, and have the negative x -axis as an asymptote (as $x \rightarrow -\infty$, $y \rightarrow 0$ but never reaches it). $y = \ln x$ is the **reflection** of $y = e^x$ in the line $y = x$ (since $\ln$ is the inverse of e^x — inverse functions are always mirror images of each other across $y = x$): it passes through $(1, 0)$, is only defined for $x > 0$, and has the y -axis ($x = 0$) as a vertical asymptote.

Sketching these graphs from the key features above, then applying the transformation rules from 2.7 as needed:

KEY PROPERTIES OF THE GRAPHS

- $y = e^x$: passes through $(0,1)$; $y > 0$ always; horizontal asymptote $y = 0$ as $x \rightarrow -\infty$; strictly increasing.
- $y = \ln x$: passes through $(1,0)$; domain $x > 0$; vertical asymptote $x = 0$; strictly increasing, but more and more slowly.
- A transformation like $y = e^{x-2} + 3$ shifts the standard e^x graph right 2 and up 3 — which also shifts its asymptote from $y=0$ to $y=3$.
- A transformation like $y = \ln(x+4)$ shifts the standard $\ln x$ graph left 4, which shifts the vertical asymptote from $x=0$ to $x=-4$, and the domain from $x>0$ to $x>-4$.

Worked example

EXAMPLE 6.3

Sketch $y = \ln(x - 1)$, stating its domain and any asymptote, and state the coordinates of the point where it crosses the x-axis.

Step 1 This is $y = \ln x$ translated 1 unit to the right (M1, comparing with $f(x-a)$ from 2.7). Domain: we need $x - 1 > 0$, i.e. $x > 1$ (A1).

Step 2 Asymptote: the vertical asymptote of $\ln x$ ($x=0$) shifts to $x = 1$ (A1).

Step 3 x-intercept: $\ln x$ crosses the x-axis at $(1, 0)$, so $\ln(x-1)$ crosses at $x-1=1$, i.e. at **(2, 0)** (A1).

Answer (The sketch: a curve starting just to the right of the vertical line $x=1$, rising slowly through $(2,0)$, continuing to increase without bound but at an ever-decreasing rate.)

EXAM TIP

Sketching $\ln(x-1)$ with the same domain and asymptote as $\ln x$ (i.e. forgetting the shift changes them) — always re-derive the domain and asymptote from the transformed equation itself, rather than assuming they're unchanged.

6.4 Exponential growth and decay models

Concept

Many real processes — population growth, radioactive decay, cooling, compound interest — change at a rate proportional to their current size, and this behaviour is modelled by an exponential function $N = N_0 e^{kt}$, where N_0 is the initial value, t is time, and k is a constant (positive for growth, negative for decay). This is the natural, calculus-based counterpart of the geometric sequences in 4.2 — the same underlying idea (constant proportional change), but modelled continuously rather than in discrete steps.

Procedure

Given a model $N = N_0 e^{kt}$:

1. N_0 is found by substituting $t = 0$ (since $e^0 = 1$, $N(0) = N_0$ directly).
2. k is found using one other known data point, then solving the resulting exponential equation (as in 6.2).
3. To find when N reaches a target value, or how long until it doubles/halves, set up and solve $N_0 e^{kt} = (\text{target})$, taking logs.
4. The **doubling time** for a growth model works out to $\ln 2/k$, and the **half-life** for a decay model (k negative) is $\ln 2/|k|$ — worth recognising as a shortcut, though you can always derive it from scratch.

Worked example

EXAMPLE 6.4

A population of bacteria is modelled by $N = 200e^{(0.05t)}$, where N is the number of bacteria and t is time in hours. (a) Find the initial population. (b) Find the population after 10 hours, to the nearest whole number. (c) Find how long it takes for the population to double.

Step 1 (a) At $t=0$: $N = 200e^0 = 200 \times 1 = \mathbf{200}$ (B1).

Step 2 (b) At $t=10$: $N = 200e^{(0.05 \times 10)} = 200e^{0.5} = \mathbf{330}$ (nearest whole number, since $200e^{0.5} = 329.74\dots$) (M1, A1).

Step 3 (c) We need $N = 2 \times 200 = 400$: $200e^{(0.05t)} = 400 \implies e^{(0.05t)} = 2 \implies 0.05t = \ln 2$ (M1, taking $\ln$ of both sides).

Answer $t = \ln 2 / 0.05 = \mathbf{13.9 \text{ hours}}$ (3 s.f.) (A1).

EXAM TIP

Substituting the doubled value into the model incorrectly — e.g. trying to solve $200e^{(0.05t)} = 200 + 2$ rather than $200e^{(0.05t)} = 2 \times 200$; “double” means multiply by 2, not add 2, and this distinction matters a great deal for exponential models.

6.5 Reducing a relationship to linear form

Concept

If you suspect two variables are related by a power law ($y = ax^n$) or an exponential law ($y = ab^x$), but don't know the constants, taking logs of both sides turns either relationship into a **straight line** — and once it's linear, you can read the constants off the gradient and intercept, exactly as you would for any straight-line graph (as in 3.1's coordinate geometry). This is a genuinely important real-world technique: scientists plot experimental data in log form specifically to test whether a power or exponential model fits, and to extract the model's constants from the fit.

The two forms you need to recognise:

POWER LAW AND EXPONENTIAL LAW

- **Power law** $y = ax^n$: taking logs (any consistent base) gives $\log y = \log a + n \log x$. Plotting $\log y$ against $\log x$ gives a straight line with **gradient n** and **y-intercept $\log a$** (so $a = 10^{(\text{intercept})}$, or $e^{(\text{intercept})}$ if using $\ln$).
- **Exponential law** $y = ab^x$: taking logs gives $\log y = \log a + x \log b$. Plotting $\log y$ against x (not $\log x$ this time) gives a straight line with **gradient $\log b$** and **y-intercept $\log a$** .
- The key skill is recognising which of the two forms applies, based on which axis (x or $\log x$) the data is plotted against, and correctly converting the gradient/intercept back into the original constants a and n (or b).

Worked example

EXAMPLE 6.5

Experimental data is believed to follow a power law $y = ax^n$. When $\ln y$ is plotted against $\ln x$, the resulting straight line has gradient 2 and passes through the point $(\ln 2, \ln 12)$. Find a and n , and hence write down the relationship between y and x .

Step 1 Since $\ln y = \ln a + n \ln x$ is a straight line in $(\ln x, \ln y)$, and the gradient is n , we immediately have **$n = 2$** (B1, direct from the gradient).

Step 2 Substitute the given point into $\ln y = \ln a + n \ln x$: $\ln 12 = \ln a + 2 \ln 2$ (M1).

Step 3 $\ln a = \ln 12 - 2 \ln 2 = \ln 12 - \ln 4 = \ln(12/4) = \ln 3$ (M1, using the laws of logs from 6.1 in reverse). So **$a = 3$** (A1).

Answer The relationship is **$y = 3x^2$** (A1).

EXAM TIP

Mixing up which variable goes with which law — plotting $\log y$ against x (not $\log x$) tests for an *exponential* relationship, while plotting $\log y$ against $\log x$ tests for a *power* relationship; using the wrong pair of axes means the gradient and intercept don't mean what you think they mean.

Chapter summary

Key formulas

KEY FORMULAS

- **Laws of logarithms:** $\log_a(xy) = \log_a x + \log_a y$; $\log_a(x/y) = \log_a x - \log_a y$; $\log_a(x^n) = n \log_a x$
- **Change of base:** $\log_a x = \ln x / \ln a$
- **Growth/decay:** $N = N_0 e^{(kt)}$; doubling time $= \ln 2/k$; half-life $= \ln 2/|k|$
- **Power law:** $y = ax^n \implies \ln y = \ln a + n \ln x$ (plot $\ln y$ vs $\ln x$)
- **Exponential law:** $y = ab^x \implies \ln y = \ln a + x \ln b$ (plot $\ln y$ vs x)

Common exam mistakes across this chapter

- Forgetting to check solutions of log equations against the domain (argument of every log must be > 0).
- Sign or arithmetic slips when combining/splitting log expressions — write out each law explicitly rather than doing it in your head.
- Confusing “double/halve” (multiply/divide by 2) with “add/subtract 2” in growth and decay word problems.
- Mixing up which axis pairing ($\log y$ vs x , or $\log y$ vs $\log x$) corresponds to which type of model when reducing to linear form.

You should now be able to

- Apply the laws of logarithms and solve exponential and logarithmic equations, checking domains

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- Sketch and transform the graphs of a^x , e^x and $\ln x$, including their asymptotes
 - Set up and solve an exponential growth or decay model, including doubling time and half-life
 - Reduce a power or exponential relationship to linear form to find unknown constants from a graph